



Agentic AI Workshop 2026 • Capstone Project 2 of 5

BUSINESS PERFORMANCE ANALYST AGENT

Ask the numbers a question in plain English. Get the answer,
the chart and the reason why.



University
of Essex

CP-02 • 6:00
1-2 October 2026



THE PROBLEM

“HOW DID WE DO LAST MONTH?”

“WHICH PRODUCTS ARE LOSING MARGIN?”

DAYS IN A QUEUE

Simple questions wait for an analyst to be free.

A SPREADSHEET BACK

The answer arrives needing its own explanation.

REPEAT REPORTING

Analysts spend their week on it instead of deeper work.



WHAT THE AGENT DOES

01

MAP THE QUESTION

to the right tables and metrics: revenue, margin, volume, cost

02

WRITE AND RUN SQL OR PANDAS

and fix its own errors from the hint each tool returns

03

DRAW THE RIGHT CHART

bar, line or breakdown, chosen to suit the question

04

EXPLAIN WHAT DROVE IT

volume, price, mix, channel and category, before giving a reason

05

FLAG ANOMALIES

months or segments well off their own recent trend

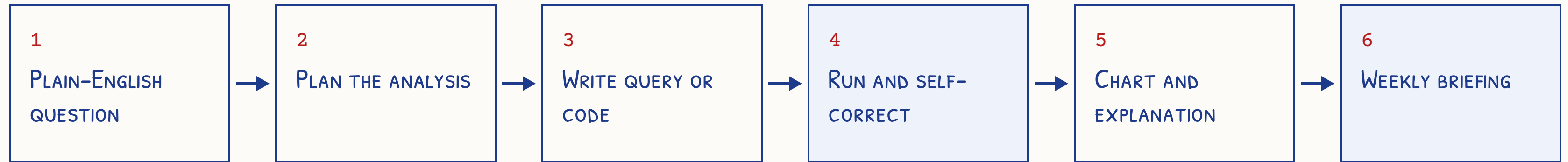
06

SHOW ITS WORKING

the exact queries, taken from the tool log



How IT WORKS



SAME ENGINE, REAL DATA

We kept the course notebook's agent loop and pointed it at Microsoft's AdventureWorks sample: 31,465 orders, May 2022 to June 2025, ten territories.

KEEPS GOING

When Groq's free daily allowance runs out, it switches to OpenAI automatically. Shorter prompts cut the cost per round by 32%.



FIVE TOOLS, PLUS ONE

Tool	PURPOSE	WHAT WE ADDED
<code>get_schema()</code>	Tables, columns, data dictionary	Also lists what the data does not have
<code>run_sql(query)</code>	Read-only query execution	One SELECT at a time; file opened read-only
<code>run_python(code)</code>	pandas analysis in a sandbox	No imports, files or system; 10-second limit
<code>make_chart(data, type)</code>	Bar, line or breakdown chart	Drawn from any query result
<code>detect_anomalies(metric)</code>	Flags values off trend	7 metrics, by territory, category, channel
<code>explain_change(...)</code>	Stretch: volume, price, mix	Three parts that always add up



BUILT TO BE TRUSTED

READ-ONLY DATABASE

The file is opened read-only, and only SELECT is allowed. "Delete the orders" is refused.

ALWAYS SHOW THE QUERY

The queries under each answer come from the tool log, never from the AI's own words.

SAY WHEN THE DATA CAN'T ANSWER

Returns, competitor prices, satisfaction: not in the data, so it says so. No stand-ins.

NEVER A FORECAST AS FACT

It shows the past trend. Any projection is labelled "ESTIMATE, not a fact".



WHY DID NORTHWEST MARGIN DROP IN Q2 2023?

6:00
CP-02
1-2 Oct 26

BUSINESS PERFORMANCE ANALYST AGENT

A

B

22-25

COVERWHAT CHANGEDTRACKSTOOLSBRIEFINGSCORECARDSTORY

LIVE AIREPLAYDRY RUN

0:02 of 1:00

VIEW FOR

STAKEHOLDERDEVELOPER

ONLINE REPLAY - LIVE AI RUNS LOCALLY

EXPLAIN THIS PAGE

SIDE ASIDE B

SIDE A3:40

01SALES BY TERRITORY, LAST YEAR0:40

02FOLLOW-UP: ...AND BY CATEGORY?0:30

▶WHY NORTHWEST MARGIN DROPPED, Q2 20231:00

04FASTEST-GROWING CATEGORY0:35

05ANYTHING UNUSUAL LAST QUARTER?0:55

BOTH SIDES6:00 OF 6:00

Your own questions: local app only

Ask

Follow-up questions remember the last two answers.

New conversation

WHY DID MARGIN DROP IN THE NORTHWEST IN Q2 2023?

REPLAY OF A RECORDED LIVE RUNSide A - budget 1:00

Replay: a recorded live run, played back step by step.

1Read the data dictionary: which numbers exist, and for which dates

2Split the change into volume, price and mix

3Split the change into volume, price and mix

4Split the change into volume, price and mix

5Split the change into volume, price and mix

6Pulled numbers from the sales database

7Drew a chart: Northwest margin change from Q1 2023 to Q2 2023 by channel

ANSWER

Northwest margin fell in Q2 2023 mainly because margin per unit deteriorated

-\$150,905

Margin went from \$110,755 in Q1 to -\$40,151 in Q2.

- Margin per unit: -\$256,848
- Higher volume: +\$240,225
- Reseller channel: -\$174,193
- Mountain Bikes: -\$166,669

Every figure has its query underneath.

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07



THE PLANTED ANOMALY, FOUND

—\$96,403

Northwest margin, May 2024

We planted a 25% bike discount in a copy of the data, then asked a leadership question that leads to it.

The agent traced the loss to the reseller discount, which jumped from **0.02%** to **21.69%**.

A rehearsal on a separate copy of the database. The real data is never changed.



The Weekly Leadership Briefing

6:00

CP-02

1-2 Oct 26

BUSINESS PERFORMANCE ANALYST AGENT

A

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22-25

COVERWHAT CHANGEDTRACKSTOOLSBRIEFINGSCORECARDSTORY

LIVE AI REPLAY DRY RUN0:04 of 0:40

VIEW FORSTAKEHOLDERDEVELOPER

ONLINE REPLAY - LIVE AI RUNS LOCALLY

EXPLAIN THIS PAGE

WEEKLY LEADERSHIP BRIEFING

One page: the KPI table straight from SQL, the agent's commentary and charts, and every query it used.

WEEK STARTING

2025-06-23

WRITE THE BRIEFING

AdventureWorks books Reseller orders about once a month, so weekly numbers swing a lot. The briefing says so.

REPLAY OF A RECORDED LIVE RUN

Week of 23 Jun 2025

THE WEEK STARTING 2025-06-23 REBOUNDED MODESTLY, BUT THE QUARTER STILL SHOWS A MAJOR DISRUPTION FROM MISSING RESELLER ACTIVITY AND A SHARP SWING IN MARGIN %.

KPI	THIS WEEK	LAST WEEK	VS LAST WEEK	4-WEEK AVG	VS 4-WEEK AVG
Revenue	\$12,388	\$11,667	+6.2%	\$71,194	-82.6%
Margin	\$6,879	\$6,461	+6.5%	\$30,877	-77.7%

KPIs STRAIGHT FROM SQL

So the AI can't mistype a number.

AI WRITES THE STORY

Commentary and charts on what moved.

ONE PAGE

With every query it used listed.



SCORECARD

7 / 7

The brief’s four questions plus three guardrail traps, marked against answers worked out from the database.

It was 6 of 7 until we fixed how it answered “return rate”.
Plus 27 automatic checks that need no AI key.

6:00

CP-02

1-2 Oct 26

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22-25

COVERWHAT CHANGEDTRACKSTOOLSBRIEFINGSCORECARDSTORY

LIVE AIREPLAYDRY RUN0:02 of 0:30

VIEW FORSTAKEHOLDERDEVELOPER

ONLINE REPLAY · LIVE AI RUNS LOCALLY

EXPLAIN THIS PAGE

SCORECARD

The brief's four questions plus three guardrail traps, marked against answers worked out from the database every time.

MARK THE ANSWERSREPLAY OF A RECORDED LIVE RUN

QUESTIONS

7 OF 7 PASSED

T1

WHAT WERE TOTAL SALES BY TERRITORY LAST YEAR?

✓ PASS

Last year = 2024. Top: Southwest \$9,121,932; total \$43,671,890.

all checks passed: mentions 'southwest'; number 9,121,932 matches expected 9,121,932; used run_sql

T2

WHY DID MARGIN DROP IN THE NORTHWEST IN Q2?

✓ PASS

Q2 margin fell in: 2023 (10.9% -> -3.1%), 2024 (4.0% -> 2.2%). A good answer says which year, and names a driver found with a query.

all checks passed: mentions 'northwest'; mentions 'reseller'; number 1.4 matches expected 1.81112; used explain_change

T3

WHICH PRODUCT CATEGORY IS GROWING FASTEST?

✓ PASS

2024 vs 2023: Accessories grew 575% (from a small base). Comparing the latest half-years instead is also fair, so the number check is lenient.



WHAT WE LEARNED

01	KEEP THE ENGINE, CHANGE THE INPUTS	We added a new section instead of rewriting, so nothing that worked broke.
02	NEVER TRUST A MODEL WITH NUMBERS IT CAN LOOK UP	KPI tables and query lists come from code and the tool log.
03	MEASURE THE TOKENS	On a free plan, the fixed text sent every round is the real cost.
04	PLAN FOR A LIVE DEMO TO FAIL	Labelled replays of real runs kept the demo going.
05	CHECK AGAINST THE BRIEF, NOT JUST THE TESTS	Reading it line by line found three gaps the tests missed.



THANK YOU. QUESTIONS?

Try the online replay, or run Live AI locally with your own key:

victorsaly.github.io/business-analyst-agent-demo

github.com/victorsaly/business-analyst-agent-demo



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A course demonstration. AdventureWorks is Microsoft's public sample of a made-up bicycle company; no real business data is used.